

NCAN • WADSWORTH CENTER • SUMMER 2026

Can the brain tell us how big the reflex will be?

Machine learning on twenty-year-old paired muscle and brain recordings.

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What the animals are doing

1

The reflex

A small shock to the leg nerve produces a reflex twitch in the soleus muscle. Its size is what we measure.

2

The training

The rat is rewarded whenever the reflex is bigger than a threshold — or smaller, depending on the group.

3

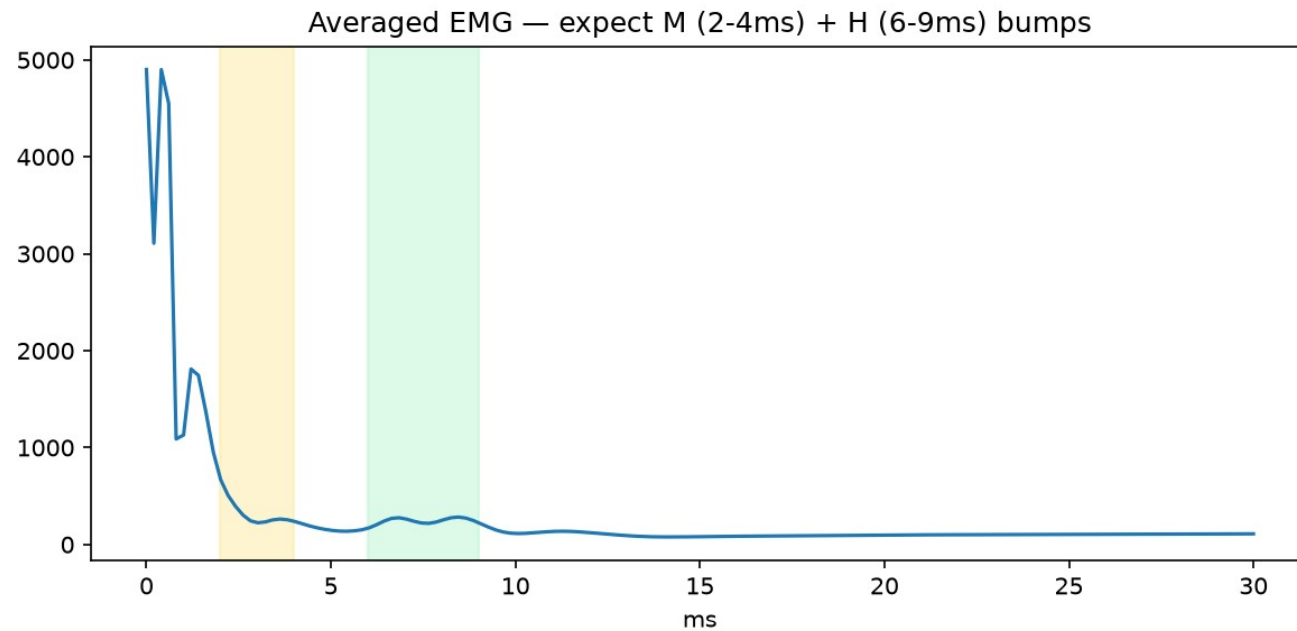
The learning

Over weeks the reflex shifts in the trained direction. A 20% change counts as success. The change is in the spinal cord.

Known already: cortical activity influences reflex size (Boulay, Chen & Wolpaw 2015). **Our question:** is that visible on a single trial, and does it change as the animal learns?

What a single trial looks like

Every shock produces one trial: a 30-millisecond window, recorded on two electrodes at the same moment.



Reading the trace

- **The spike at zero**

the stimulus itself, an electrical artifact

- **M-wave, 2-4 ms**

the nerve firing the muscle directly

- **H-reflex, 6-9 ms**

the signal that travelled to the spinal cord and back — this is what conditioning changes

Six animals, 2.7 million trials

2.7M

trials decoded

6

animals

3 / 3

down / up

45–210

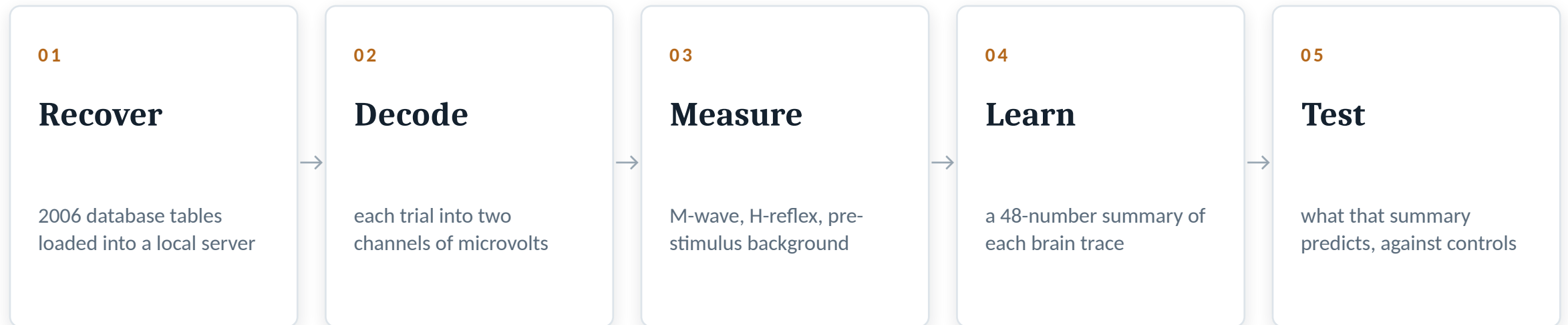
days each

Finding the animals that actually learned — without downloading everything

The log files are tiny. The experimenter raises the reward threshold whenever the rat starts beating it, so the threshold history is a record of how the animal was doing. We read only the logs, then downloaded the signal data for the animals that had learned.

One animal excluded. A12's pre-stimulus cortical signal is 2.8 μV against 106–128 μV in the others — that electrode was dead.

How the data becomes a model



The step that made it possible

The archived decoder documents these files as big-endian. They are not. Decoded as documented the signal is noise; decoded correctly the M-wave and H-reflex appear exactly where the experimenters' notes said they would.

Why the reflex size alone tells you nothing

The problem

A bigger shock makes a bigger reflex — no learning required. Across these recordings the delivered stimulus roughly quadrupled.

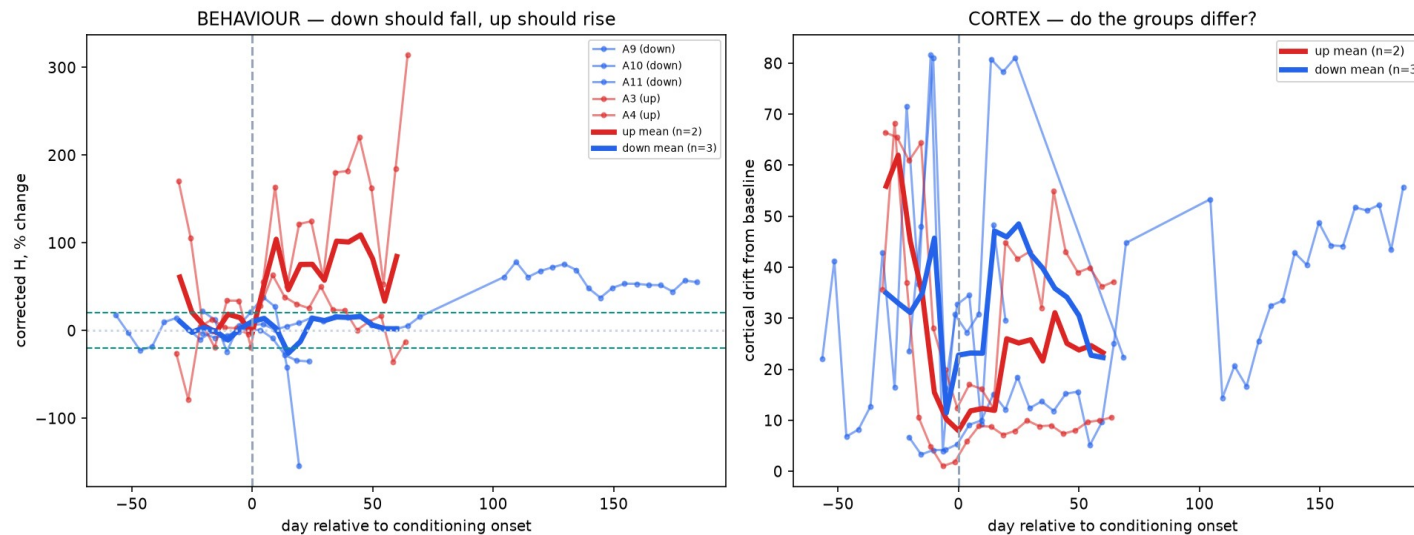
Early on, this made one of our results come out backwards.

The fix

The M-wave is in the same trace and reflects only the stimulus. It is a per-trial receipt for how much shock arrived. We remove its influence from everything.

Dr. Carp's addition: the reflex also depends on how active the muscle already was. We now measure the 20 ms before each shock and remove that too. Every number in this talk has both corrections applied.

Conditioning works, in both directions



POSITIVE CONTROL

Each animal is lined up to its own training start day. Up-trained animals (red) rise; down-trained (blue) fall.

Three of five clear the 20% criterion in the trained direction.

The two that don't become a built-in control later.

Two things change slowly over weeks

This is the central difficulty, and it took us a while to see it clearly.

Learning

The animal's reflex shifts gradually across weeks of training.

Electrode drift

A chronically implanted electrode also changes gradually across the same weeks.

Inside one animal, both are smooth functions of time — so no amount of analysis can tell them apart.

That is a limitation of the study design, not of the data. It shaped everything we did next.

Our first approach did not survive its controls

We measured how far the average brain state moved across training. It moves — but three tests say the movement is the recording, not the animal.

Take baseline-only data and pretend half of it was training

Nothing happened, so it should show no movement

It moves just as much — 5 of 6 animals

Shuffle which trial belongs to which day

Keeps the numbers, destroys the timeline

No movement — so the measure itself is sound

Compare animals trained up against trained down

Electrode drift doesn't care which way you trained

No difference between the groups

Two of these three were Dr. Carp's suggestions. They changed our conclusion.

Ask a question drift cannot fake

Instead of

“Did the average brain state move over weeks?”

We ask

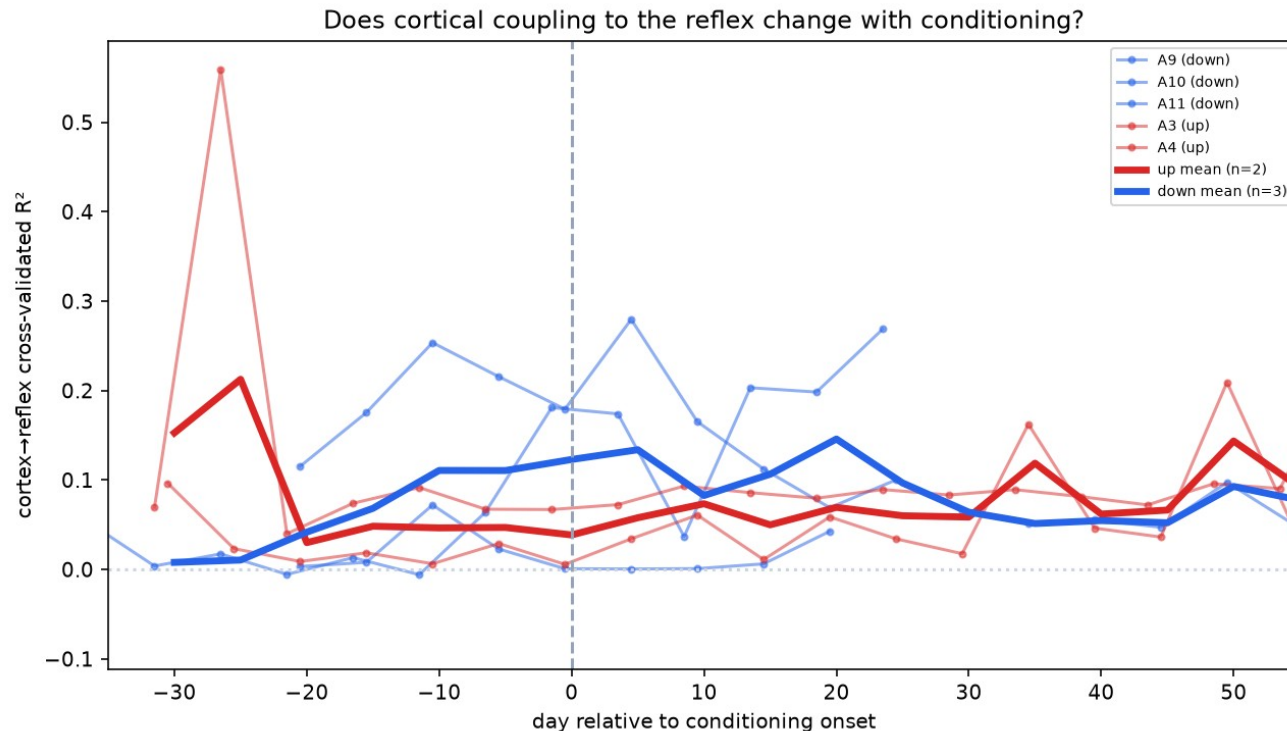
“Can the brain predict this trial's reflex?”

Why drift cannot produce this

We train and test the prediction inside a single five-day window. A slowly drifting electrode moves the training trials and the test trials together, so it cancels. Drift can shift an average. It cannot invent a trial-by-trial relationship inside five days.

And within each window we still remove the stimulus and the background first, then compare against a shuffle of that same window's data.

The brain does carry a signal about the reflex



99 of 101

five-day windows where the real data beat its own shuffled version

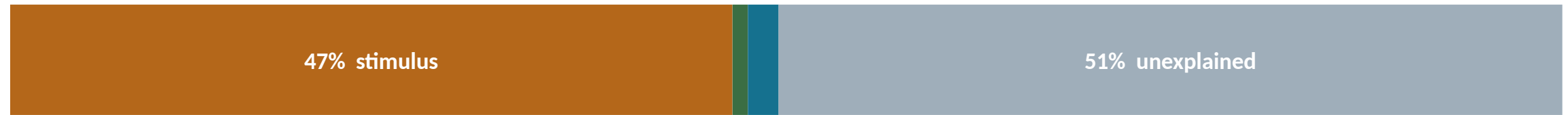
Present in all five animals.

Counting the animal as the unit of analysis:
 $t(4) = 3.81$, $p = 0.019$

Small in size, but almost never absent.

How much does the brain actually explain?

Dr. Carp asked us to split the variance up. The stimulus dominates, and the brain overlaps with it heavily.



Stimulus, from the M-wave

47%

How much shock actually arrived



Pre-stimulus background

1%

How active the muscle already was



Brain, uniquely

2%

What it adds once the other two are known



Unexplained

51%

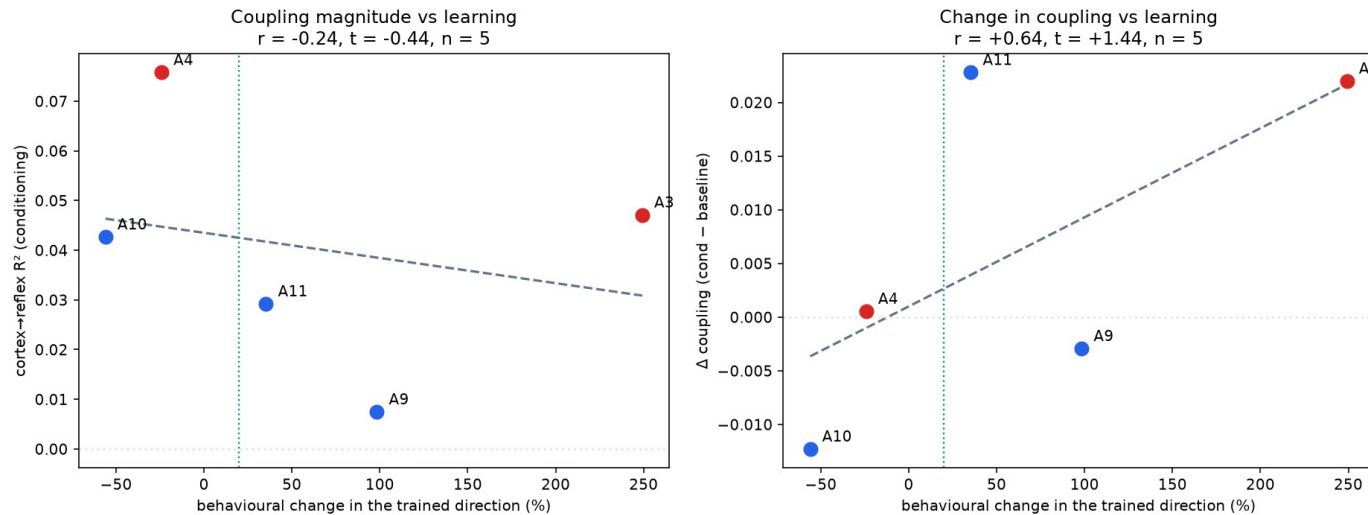
Trial-to-trial variability we can't yet account for

Everything we tried to kill it with

Baseline sham	Pretend part of the baseline was training	drift fails
Shuffled trial order	Keep the numbers, destroy the timeline	null ≈ 0
Stimulus strength	Is it just how hard we shocked?	survives
Pre-stimulus background	Is it muscle tone rather than brain?	survives
Within-window testing	Could electrode drift manufacture it?	drift cancels
Muscle crosstalk	Is the brain electrode picking up muscle?	$r = +0.06$
Up versus down	Does it depend on training direction?	no difference

The two that failed are the ones that told us the average-state approach was the wrong question.

The test that more animals would settle



$$r = +0.71$$

between how much an animal learned and how much its brain coupling changed

The two animals that failed to learn show no increase — a built-in negative control.

With five animals this is a direction, not a result. Five more are ready to process.

WHERE WE'D VALUE YOUR INPUT

Four questions

1

Is a 2% unique contribution worth reporting?

It is small, but consistent, and it survives every control we could think of.

2

Does the within-window design convince you?

Train and test inside five days so drift cancels. Is that enough to rule out the electrode?

3

What else could produce this?

We've excluded stimulus, muscle tone, drift and crosstalk. What are we missing?

4

Is the negative result worth publishing?

The obvious analysis looks convincing and fails three controls. Useful warning, or a distraction?